



PANYNJ DBE | VIRGINIA-CERTIFIED DBE / MBE

PRINCIPAL-LED STRUCTURAL & BRIDGE ENGINEERING

Peng Chen, P.E. | Founder & Principal

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CORE SERVICES

- Bridge and transportation-structure inspection
- Structural condition assessment and deficiency documentation
- Bridge design, rehabilitation and repair detailing
- Load rating and structural capacity evaluation
- Building, parking-structure and facility assessment
- Construction engineering, RFIs and field-design support
- Temporary works and contractor-oriented solutions

CREDENTIALS

- Professional Engineer: VA, MD, DC, NY, NJ and WA
- FHWA NBIS Certified Bridge Inspector
- M.S., Civil Engineering - Rutgers University
- 20+ years of bridge and infrastructure experience

TECHNICAL RESOURCES

AASHTOWare BrR | CSI Bridge | MIDAS | LARSA |
SAP2000 | STAAD | LEAP Bridge | AutoCAD | MicroStation
| Mathcad

CODES & STANDARDS

AASHTO | AREMA | FHWA | ACI | AISC | ASCE 7 | IBC |
DOT criteria and specifications

PROCUREMENT PROFILE

Legal Name	Polaris Bridge Engineering LLC
Firm Type	Virginia limited liability company
Status	PANYNJ DBE Virginia DBE / MBE
NAICS	541330 - Engineering Services

COMPANY PROFILE

Polaris Bridge Engineering LLC provides principal-led bridge and structural engineering support to prime consultants, contractors and public owners. The firm combines senior-level technical judgment with responsive, practical delivery for inspection, rehabilitation, load-rating, design and construction-phase assignments.

PANYNJ-RELEVANT EXPERIENCE

Representative prior professional experience of Peng Chen, P.E. before forming Polaris:

- Structural inspection Team Leader for aviation and marine-terminal facilities at JFK, Newark Liberty, LaGuardia, Port Newark and Port Elizabeth.
- Inspection experience involving airport terminals, multi-level parking garages, hangars, high-mast light towers and supporting facility structures.
- Condition documentation, structural-deficiency evaluation and development of practical repair recommendations for steel and concrete structures.

SELECTED POLARIS ASSIGNMENTS

- Facility sinkhole investigation and engineering repair recommendations, including assessment of drainage-related subsurface distress.
- Structural evaluation and support design for rooftop mechanical equipment and associated steel framing.
- Commercial and institutional structural condition investigations, repair concepts and retrofit support.

REPRESENTATIVE INFRASTRUCTURE EXPERIENCE

- Bridge and structural engineering support for the Hampton Roads Bridge-Tunnel Expansion design-build program.
- Structural and construction-engineering experience on New York's East Side Access transit program.
- Bridge design, rehabilitation and load-rating experience across VDOT, NYSDOT, NJDOT and other DOT programs.

WHY POLARIS

- Direct principal involvement from scoping through final deliverables
- Deep familiarity with bridges, transit, airports and operational facilities
- Focused DBE subconsultant support that integrates with larger project teams
- Practical recommendations grounded in inspection, design and construction experience

Available for teaming and subconsultant assignments

Bridge | Transit | Airport | Highway | Building & Facility Structures